

HMF Design Clinic Solution Guide

Reference answers and the three assessment dimensions

This is your model-answer companion. For every pain point you will find the fitting Cisco solution, how a strong Solution Engineer explains it, and the outcome it delivers, mapped to the three dimensions so you can assess answers consistently.

The scenario. Cedarline Health Group is a regional healthcare payer-provider with roughly 8,000 employees across four hospitals and thirty-one clinics, plus a claims-processing business that handles the money. Growth by acquisition (including a second data centre absorbed as-is) and a mid-flight, disorderly move to public cloud have left a fragmented security posture. After a security incident in March, the board asked what would have stopped it, and leadership still cannot answer. Budget and executive attention are approved but explicitly temporary, and the security team is small.

Score each pain point across three dimensions

Each pain point is scored across the three dimensions below: Problem Identifier out of 5, Solution Designer out of 10, and Value Communicator out of 10, for 25 per pain point. Four pain points make a case study score out of 100, and the four case studies make an overall score out of 400. The model answers that follow map to these three dimensions. Use the Grading Guide for the full rubric.

Problem Identifier OUT OF 5

Correctly identified the customer's pain point and the requirements behind it.

Solution Designer OUT OF 10

Recommended the right Cisco solution, architecture, and approach.

Value Communicator OUT OF 10

Clearly explained how the solution works and the customer outcomes it enables.

What full credit looks like. Problem Identifier: names the underlying requirement and constraints, not just the surface ask. **Solution Designer:** recommends the right Cisco solution and an architecture or approach that fits Cedarline. **Value Communicator:** explains clearly how it works and states the concrete Cedarline outcome. Reward reasoning specific to Cedarline over generic best practice.

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Case Study 1

The conversation. Cedarline is standing up AI across regulated workflows and public cloud faster than security can see it. Help them prove compliance, vet what they adopt, and get visibility and guardrails over AI traffic.

PAIN POINT 1

Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?

Problem Identifier	The team should recognise: Provider-level evidence of robustness and cybersecurity under EU AI Act Article 15.
Solution Designer	Cisco AI Defense (AI Model and Application Validation)
Value Communicator	How it works. Cisco AI Defense is specifically designed to support Provider obligations under the EU AI Act. We utilize algorithmic red teaming technology for AI Model & App Validation, which identifies safety and security vulnerabilities at scale. Outcome. It automates the technical evidence gathering required for "Conformity Assessments." By mapping AI-specific threats to standardized frameworks (like MITRE ATLAS and OWASP), it provides the documentation and continuous monitoring necessary to prove to regulators that your systems are robust and compliant.

PAIN POINT 2

We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?

Problem Identifier	The team should recognise: Vetting AI models and third-party agents for vulnerabilities before they reach production.
Solution Designer	Cisco AI Defense (Supply Chain Risk Management)
Value Communicator	How it works. Cisco AI Defense addresses this through proactive Supply Chain Risk Management. We scan your model files, repositories, and MCP servers to identify malicious or unsafe assets. Outcome. It enables "Shift-Left" security by creating an inventory of all AI models and agents. By identifying vulnerabilities before production, it prevents the introduction of malicious code or poisoned data into your ecosystem.

PAIN POINT 3

We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.

Problem Identifier	The team should recognise: Visibility and consistent control over AI assets (models, agents, datasets and MCP tools) across the cloud.
Solution Designer	Cisco AI Defense (AI Cloud Visibility)
Value Communicator	How it works. That's exactly the visibility gap Cisco AI Defense is built to solve. With AI Cloud Visibility, Cisco AI Defense continuously discovers AI assets across distributed cloud environments, including models, agents, datasets, MCP servers, and agent-to-tool workflows. Outcome. Identifies rogue or unsanctioned AI assets. Maps relationships across data, models, agents, and tools.

PAIN POINT 4

We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.

Problem Identifier	The team should recognise: Seeing AI assets and third-party model usage, and enforcing runtime guardrails on AI traffic without changing applications.
Solution Designer	Cisco AI Defense with Cisco Multicloud Defense (AI Runtime Protection)
Value Communicator	How it works. That's exactly what the Cisco AI Defense-Cisco Multicloud Defense integration is designed to address. Cisco AI Defense uses the integration to discover AI assets in your cloud environments and identify external AI destinations your workloads are calling. Outcome. Finds AI assets in cloud accounts. Identifies third-party AI model usage.

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Case Study 2

The conversation. Workloads sprawl across AWS, Azure and on-prem, each with its own policy model. Contain lateral movement, prove consistent controls to auditors, and give Cedarline one place to define and enforce policy.

PAIN POINT 1

If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?

Problem Identifier	The team should recognise: Containing lateral movement between public cloud and on-prem in a multicloud environment.
Solution Designer	Cisco Multicloud Defense (macro and micro-segmentation gateways)
Value Communicator	How it works. Cisco Multicloud Defense leverages macro and micro-segmentation capabilities to enforce strict communication boundaries. By integrating with our Hybrid Mesh Firewall and Hypershield, we can isolate workloads and inspect traffic between cloud segments, effectively stopping lateral movement before it reaches your sensitive data. Outcome. It provides granular visibility and control over east-west traffic. By applying Zero Trust principles across the hybrid fabric, you transform your network into a series of secure, isolated zones that prevent an attacker from pivoting through your infrastructure.

PAIN POINT 2

Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.

Problem Identifier	The team should recognise: Consistent, auditable proof that security controls are applied across all cloud workloads.
Solution Designer	Cisco Multicloud Defense (unified logging and visibility)
Value Communicator	How it works. Cisco Multicloud Defense provides unified logging and traceability. Because our security framework is integrated across the network fabric, you can generate consolidated reports that map directly to your compliance requirements (such as NIS2 or DORA) across all cloud environments from a single dashboard. Outcome. It automates the collection of evidence for security audits. By consolidating logs and policy enforcement data, you create a "single source of truth" that simplifies compliance reporting and demonstrates proactive governance to regulators.

PAIN POINT 3

We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.

Problem Identifier	The team should recognise: One consistent policy model across AWS, Azure and on-prem, without per-cloud overhead.
Solution Designer	Cisco Multicloud Defense (unified policy across public clouds)
Value Communicator	How it works. Cisco Multicloud Defense, integrated within our Hybrid Mesh Firewall architecture, provides a unified control plane. You can define security policies once via Security Cloud Control and enforce them consistently across your entire distributed environment, regardless of the underlying cloud provider. Outcome. It replaces siloed, manual configuration with centralized orchestration. By abstracting the security policy from the cloud-native tools, it ensures that your security posture is uniform and eliminates the "configuration drift" that occurs when managing multiple cloud consoles.

PAIN POINT 4

We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.

Problem Identifier	The team should recognise: Unified segmentation that prevents policy drift across hybrid and multicloud.
Solution Designer	Cisco Secure Workload with Cisco Secure Firewall (managed via FMC and Security Cloud Control)
Value Communicator	How it works. That's a common hybrid environment challenge. Cisco Secure Workload helps create a more unified segmentation model by using application dependency awareness and AI/ML-driven policy discovery to define policy based on how workloads actually communicate. Outcome. Reduces policy drift across hybrid and multicloud environments. Aligns policy to application behavior, not just static IP-based rules.

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Case Study 3

The conversation. A mixed-vendor perimeter, an inherited rulebase and mostly encrypted traffic leave blind spots and a slow response. Restore visibility, speed the response, and modernise detection with the team they have.

PAIN POINT 1

When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.

Problem Identifier	The team should recognise: Automated, real-time isolation of a compromised device, faster than a manual response.
Solution Designer	Cisco ISE (Security Group Tags via pxGrid and Adaptive Network Control)
Value Communicator	How it works. Cisco ISE integrates with your security ecosystem via pxGrid and Adaptive Network Control (ANC). When a threat is detected, ISE can automatically trigger a quarantine policy, isolating the compromised device in real-time without requiring manual intervention. Outcome. It automates the "Response" phase of the incident lifecycle. By bridging the gap between detection and enforcement, it stops threats at the speed of the network.

PAIN POINT 2

Our firewall environment has become too complex to manage manually.

Problem Identifier	The team should recognise: Simplifying and cleaning up firewall policy administration at scale.
Solution Designer	Cisco Security Cloud Control with FMC and Cisco Secure Firewall (unified policy administration)
Value Communicator	How it works. Cisco applies AI-driven analysis to firewall telemetry and policy data to surface optimization opportunities, reduce manual troubleshooting, and guide cleaner policy administration. Outcome. Reduces manual policy review; identifies redundant/stale rules; lowers risk from human error.

PAIN POINT 3

Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.

Problem Identifier	The team should recognise: Restoring visibility into encrypted traffic without decrypting everything.
Solution Designer	Cisco Secure Firewall with the Encrypted Visibility Engine (EVE)
Value Communicator	How it works. Encrypted Visibility Engine (EVE) uses AI/ML-based fingerprinting to analyze encrypted traffic patterns and identify suspicious behavior without decrypting the payload. Outcome. Restores visibility into encrypted blind spots; avoids the performance/privacy overhead of decrypt-everything strategies.

PAIN POINT 4

We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.

Problem Identifier	The team should recognise: Detecting unknown and zero-day threats beyond signature-based detection.
Solution Designer	Cisco Secure Firewall with Snort ML
Value Communicator	How it works. That's exactly where Snort ML helps. It adds machine-learning-based detection inside the Snort 3 engine to identify exploit behavior associated with unknown/zero-day threats. Outcome. Addresses the gap between known and unknown threat detection; reduces dependence on signature timing.

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Case Study 4

The conversation. A flat data centre and fragmented Kubernetes networking invite east-west movement. Build segmentation Cedarline can adopt safely, with runtime visibility and application-dependency awareness so nothing breaks.

PAIN POINT 1

Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.

Problem Identifier	The team should recognise: Runtime detection of suspicious behaviour and policy violations inside workloads.
Solution Designer	Isovalent Tetragon (eBPF runtime security)
Value Communicator	How it works. That's where Tetragon comes in. Tetragon extends the value of Cilium by providing eBPF-based runtime security and threat detection. Outcome. Adds runtime visibility beyond basic network controls. Detects suspicious system and network behavior in real time.

PAIN POINT 2

Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.

Problem Identifier	The team should recognise: Consistent Kubernetes networking and policy across clusters, with less operational complexity.
Solution Designer	Isovalent Cilium (eBPF Kubernetes networking and policy)
Value Communicator	How it works. That's exactly where Isovalent Cilium helps. Cilium provides a modern, eBPF-based Kubernetes networking layer that gives you consistent connectivity, policy enforcement, and visibility across clusters and environments. Outcome. Replaces brittle, IP-centric networking models with a Kubernetes-native approach. Improves consistency across dynamic cluster environments.

PAIN POINT 3

Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.

Problem Identifier	The team should recognise: Consolidating Kubernetes networking, security and observability to reduce tool sprawl.
Solution Designer	Isovalent Cilium with Tetragon
Value Communicator	How it works. That's exactly the consolidation opportunity Isovalent addresses. With Cilium for networking and policy, and Tetragon for runtime security, you get a more unified platform for connectivity, segmentation, observability, and security. Outcome. Consolidates multiple functions into one platform approach. Reduces tool sprawl and operational friction.

PAIN POINT 4

We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.

Problem Identifier	The team should recognise: Mapping east-west flows and application dependencies to segment safely without breaking applications.
Solution Designer	Cisco Secure Workload (application dependency mapping)
Value Communicator	How it works. That's one of the biggest reasons customers adopt Cisco Secure Workload. It gives you deep visibility into workload communication and application dependency mapping, so you can see how applications actually behave before enforcing policy. Outcome. Reveals actual application communication paths. Reduces the risk of segmentation causing outages.